Patterns in the Times Table

Look for patterns. They help you remember the facts.

The times table

×	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

Row 4 is all even:
4 × anything is even.

3 × 7 and 7 × 3 match:
turnarounds mirror across the table.

Going down a column:
add the top number each time.

Odd × odd is odd:
3 × 5 = 15, 7 × 9 = 63.

Double a number, double the answer

$3 \times 6 = 18$

$6 \times 6 = 36$

6 is double 3,
so 36 is double 18.

Your turn!

×4 4 8 12 16 20 24

For teachers

Goal: find patterns in the times table and explain them (3.OA.9). Ask "Why is every number in the 4s row even?" and "What do you notice about 3 × 7 and 7 × 3?" Patterns make the hard facts easier to remember.

Patterns in the Times Table: Practice

Use the patterns. Check with the table if you need to.

Part 1 Fill in the row Add the same number each time.

1)	×3	3	6	9	12	15	18	21	24	27	30
2)	×6	6	12	18	24	30	36	42	48	54	60
3)	×8	8	16	24	32	40	48	56	64	72	80
4)	×7	7	14	21	28	35	42	49	56	63	70
5)	×9	9	18	27	36	45	54	63	72	81	90
6)	×4	4	8	12	16	20	24	28	32	36	40

Part 2 Pieces of the table Fill in the missing products.

7)

×	4	5	6	7
3	12	15	18	21
4	16	20	24	28
5	20	25	30	35

8)

×	2	3	4	5
6	12	18	24	30
7	14	21	28	35
8	16	24	32	40

Patterns in the Times Table: Practice (continued)

Part 2 Pieces of the table (continued)

9)

×	6	7	8	9
5	30	35	40	45
6	36	42	48	54
7	42	49	56	63

10)

×	5	6	7	8
7	35	42	49	56
8	40	48	56	64
9	45	54	63	72

Part 3 Even or odd? Even × any number is even. Odd × odd is odd.

11) 5 × 7 — even **odd**

12) 4 × 7 — **even** odd

13) 7 × 7 — even **odd**

14) 3 × 8 — **even** odd

15) 5 × 5 — even **odd**

16) 9 × 3 — even **odd**

17) 8 × 3 — **even** odd

18) 2 × 9 — **even** odd

19) 6 × 9 — **even** odd

Part 4 Double it No help this time. Double one number, double the answer.

20) 2 × 7 = 14 so 4 × 7 = **28**

21) 3 × 9 = 27 so 6 × 9 = **54**

22) 4 × 6 = 24 so 8 × 6 = **48**

23) 2 × 8 = 16 so 4 × 8 = **32**

24) 3 × 4 = 12 so 6 × 4 = **24**

25) 5 × 7 = 35 so 10 × 7 = **70**

Patterns in the Times Table: Practice (continued)

Part 4 Double it (continued)

26) $4 \times 9 = 36$ so $8 \times 9 =$ **72**

27) $2 \times 6 = 12$ so $4 \times 6 =$ **24**

28) $3 \times 7 = 21$ so $6 \times 7 =$ **42**

29) $4 \times 8 = 32$ so $8 \times 8 =$ **64**

Part 5 Times 9 from times 10 No help this time. Take away one group.

30) $10 \times 6 = 60$ so $9 \times 6 =$ **54**

31) $10 \times 3 = 30$ so $9 \times 3 =$ **27**

32) $10 \times 8 = 80$ so $9 \times 8 =$ **72**

33) $10 \times 4 = 40$ so $9 \times 4 =$ **36**

34) $10 \times 7 = 70$ so $9 \times 7 =$ **63**

35) $10 \times 9 = 90$ so $9 \times 9 =$ **81**

Part 6 Times 5 from times 10 No help this time. Times 5 is half of times 10.

36) $10 \times 8 = 80$ so $5 \times 8 =$ **40**

37) $10 \times 3 = 30$ so $5 \times 3 =$ **15**

38) $10 \times 6 = 60$ so $5 \times 6 =$ **30**

39) $10 \times 9 = 90$ so $5 \times 9 =$ **45**

40) $10 \times 4 = 40$ so $5 \times 4 =$ **20**

41) $10 \times 7 = 70$ so $5 \times 7 =$ **35**

Part 7 Remember? Problems from earlier packets.

42) 7 m by 6 m: area = **42** sq m

43) 40 minutes after 2:35 = **3:15**

44) Round 481 to the nearest hundred: **500**

45) $3 \times 70 =$ **210**

Patterns in the Times Table: Practice (continued)

Part 7 Remember? (continued)

46) $72 - 38 =$ **34**

47) $64 - 27 =$ **37**

Part 8 Word problems Look for a pattern.

48) Each table has 4 legs. Is the number of legs on 7 tables even or odd?

Number sentence **7 × 4 = 28, even**

Answer **even**

Draw or show your work

49) A pack has 6 juice boxes. 3 packs have 18. How many juice boxes are in 6 packs?

Number sentence **double 18 = 36**

Answer **36** juice boxes

Draw or show your work

50) 9 boxes hold 8 crayons each. How many crayons is that?

Number sentence **10 × 8 = 80, 80 − 8 = 72**

Answer **72** crayons

Draw or show your work

51) Ben counts by 5s, 8 times. What number does he say last?

Number sentence **8 × 5 = 40**

Answer **40**

Draw or show your work

Patterns in the Times Table: Practice (continued)

Part 8 Word problems (continued)

52) Challenge: Kim says 7 × 9 = 64. Odd × odd is odd, so that can't be right. What is 7 × 9?

Number sentence 7 × 9 = 63, odd

Answer 63

Draw or show your work

Check-In: Patterns in the Times Table

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Fill in the row

1) ×4 4 8 **12** **16** 20 24 28 **32** 36 **40**

Part 2 Pieces of the table

2)

×	5	6	7	8
6	30	36	42	48
7	35	42	49	56
8	40	48	56	64

3)

×	3	4	5	6
6	18	24	30	36
7	21	28	35	42
8	24	32	40	48

Part 3 Even or odd?

4) 8×9 (even) odd

5) 7×5 even (odd)

6) 5×8 (even) odd

Part 4 Double it

7) $4 \times 4 = 16$ so $8 \times 4 =$ **32**

8) $2 \times 4 = 8$ so $4 \times 4 =$ **16**